

PYTHON FOR BEGINNERS · DAY 9

Dictionaries & Tuples

Storing data with labels

Instructor: Homaira Mohammadi

```
student = {
    "name": "Sara",
    "age": 19
}
print(student["name"])
```

TODAY'S GOALS

By the end of today, you can…

- Store **key** → **value** pairs
- Add, change and delete items
- Store records (list of dictionaries)
- Get values safely with **get()**
- Loop through a dictionary
- Use **tuples** for fixed data

THE PROBLEM WITH LISTS

What was index 2 again? 🤔

List: by position

```
student = ["Sara", 19,
    "Computer Science"]

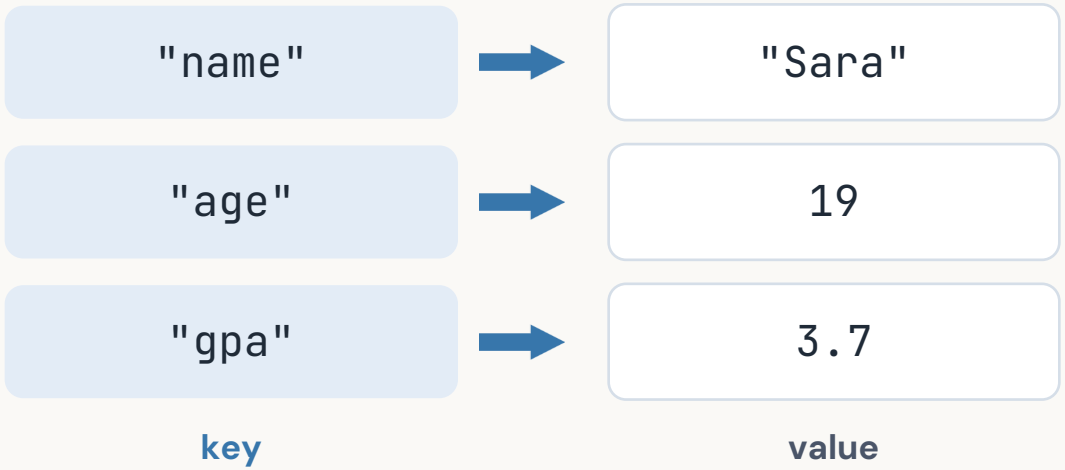
print(student[2])
```

Dictionary: by name

```
student = {
    "name": "Sara",
    "age": 19,
    "major": "CS"}
print(student["major"])
```

WHAT IS A DICTIONARY?

Pairs of key → value



Curly brackets **{ }**

Each pair is **key: value**

Keys must be **unique**

Like a real dictionary: **word → meaning**

READING VALUES

Use the key in square brackets

Code

```
s = {"name": "Sara", "age": 19}

print(s["name"])
print(len(s))
print("age" in s)
print(s.get("email", "none"))
```

Output

```
Sara
2
True
none
```

get() gives a default instead of an error.

CHANGING A DICTIONARY 🖍️

Add, change and delete

s = {"name": "Sara", "age": 19}	What it does
s["email"] = "s@mail.com"	add a new key
s["age"] = 20	change an existing key
del s["email"]	delete a key
s.pop("age")	delete and give back the value

Same code to add or change: if the key exists it's **changed**, if not it's **added**.

LOOPING 🔁

Keys and values together

Code

```
prices = {"apple": 1.5,
          "bread": 3.25,
          "milk": 4.0}

for item, price in prices.items():
    print(f"{item}: RM {price:.2f}")
```

Output

```
apple: RM 1.50
bread: RM 3.25
milk: RM 4.00
```

Also: **.keys()** and **.values()**

⚠️ WATCH OUT

A missing key = KeyError

Check with **in** first, or use **get()**.

```
s = {"name": "Sara"}

s["email"]
# ❌ KeyError

s.get("email")
# ✅ None
```

RECORDS 🗃️

A list of dictionaries

Code

```
students = [
    {"name": "Ali", "score": 85},
    {"name": "Sara", "score": 92},
    {"name": "Omar", "score": 47},
]
for s in students:
    if s["score"] >= 50:
        print(s["name"], "passed")
```

Output

```
Ali passed
Sara passed
```

Each dictionary = **one student**.
The list = **the whole class**.

TUPLES 🔒

A list that can’t be changed

```
point = (3, 5)
days = ("Mon", "Tue", "Wed")

print(days[0])  # Mon

x, y = point    # unpacking
print(x, y)     # 3 5
```

Round brackets **( )**

Indexes work like lists

But **no** append, remove or change

Use for fixed data: coordinates, days, colours

WHICH ONE?

List vs tuple vs dictionary

	List	Tuple	Dictionary
Brackets	[ ]	( )	{ }
Get items by	position	position	key
Can change?	Yes	No	Yes
Example	["Ali", "Sara"]	(3, 5)	{"name": "Ali"}
Use for	a changing collection	fixed data	labelled data

🧠 QUICK CHECK

Predict before you run!

Q1. What is printed?

```
d = {"a": 1, "b": 2}
d["c"] = 3
d["a"] = 10
print(d)
```

Q2. Error or not?

```
d = {"a": 1}
print(d.get("x"))
```

Q3. Does it work?

```
t = (1, 2, 3)
t.append(4)
```

SUMMARY

What we learned today

Code	Meaning
d = {"name": "Ali"}	create a dictionary
d["name"] · d.get("name")	read a value
d["city"] = "Kabul"	add or change
del d["city"]	delete
"name" in d	is the key there?
for k, v in d.items():	loop through pairs
point = (3, 5)	a tuple (can’t change)

💻 YOUR TURN

Practice time!

Open **exercises.ipynb** in the Day 9 folder and solve the 9 tasks, from ⭐ to ⭐ ⭐ ⭐.

⭐ Capital cities dictionary

⭐ ⭐ Shopping bill 🛒

⭐ ⭐ ⭐ Quiz from a dictionary 🧠

Next lesson: Final project 🏆